

Energy optimization of bitcoin mining integrated greenhouse with model predictive control[☆]

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HIGHLIGHTS

- Model predictive control framework for bitcoin mining integrated greenhouse.
- Waste heat from bitcoin mining reduces greenhouse heating costs.
- A case study based on bitcoin mining integrated greenhouses across global locations.
- Proposed control framework improves energy efficiency and enhances profitability.

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ABSTRACT

This study presents a novel control framework that integrates bitcoin mining waste heat with greenhouse climate regulation, leveraging model predictive control to enhance energy efficiency and economic viability. While the feasibility of using bitcoin mining waste heat for greenhouse heating through thermal modeling and parametric studies has been explored, this work focuses on real-time predictive control, dynamically regulating temperature, humidity, and CO₂ concentration based on external weather conditions and crop requirements. The MPC framework manages the dynamic and interconnected climate conditions of the greenhouse by leveraging real-time weather data and predictive models. By dynamically controlling actuators such as fans, blinds, heating systems, and CO₂ injection, the framework ensures optimal conditions for crop growth while minimizing energy costs. Simulations are conducted for greenhouses of varying sizes (individual, semi-commercial, and commercial) across eight cities representing diverse climate zones. Results show that MPC outperforms On-Off control by reducing energy consumption by up to 15 %, while maintaining climate conditions within target ranges over 95 % of the time. Furthermore, economic analysis demonstrates that integrating bitcoin mining waste heat results in significant profitability, with net annual profits reaching up to \$1.5 million for commercial-scale greenhouses under high bitcoin prices. The findings establish a scalable and intelligent control strategy for integrating bitcoin mining with agriculture, advancing sustainable food production while optimizing energy use and reducing greenhouse gas emissions in agricultural operations.

1. Introduction

The global population has surpassed 8 billion and is projected to reach 9.5 billion by 2050 [1,2], intensifying concerns about food security and resource sustainability [3,4]. Traditional agricultural practices face growing challenges, including diminishing farmland, water scarcity, and increasing energy demands, all of which threaten global food

production. At the same time, energy-intensive industries such as bitcoin mining contribute significantly to global electricity consumption and carbon emissions, raising concerns about their environmental impact. Repurposing bitcoin mining waste heat for greenhouse agriculture offers a novel approach to mitigating these issues by transforming an energy-intensive process into a resource for sustainable food production [5]. Instead of dissipating excess heat through costly cooling systems, bitcoin mining operations can provide a consistent and localized heat source for

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Nomenclature			
A_p	area of pad	q_v	heat flux from ventilation
A_s	surface area of greenhouse	T_h	water temperature in heating pipes
a_0, a_1, a_6	coefficients of CO ₂ model	T_i	indoor temperature
C_a	air specific heat	T_p	temperature of air through the cooling pad
C_w	water specific heat	T_o	outdoor temperature
H_i	relative humidity	u_{blind}	control input of shade curtain ranging from 0 to 1
h_i	absolute humidity	u_{fan}	control input of the fan speed ranging from 0 to 1
h_o	outdoor absolute humidity	$u_{fan,max}$	maximum fan speed in greenhouse
I	PAR	u_k	control inputs
I_a	outdoor global radiation	u_{light}	control input of the supplemental light ranging from 0 to 1
K_a	coefficient of solar equation	$u_{light,max}$	maximum energy input of the supplemental light system
K_c	coefficient of the cover equation	u_{pad}	control input of the pad shutter ranging from 0 to 1
K_h	coefficient of heating pipe equation	$u_{pad,max}$	maximum airflow rate through the pad
K_p	coefficient of pad cooling equation	u_{pipe}	control input of heating pipe water flow rate ranging from 0 to 1
L	leaf area index of crops	$u_{pipe,max}$	maximum water flow rate of the heating pipe
m_{fog}	water net flow from the fogging system	V	volume of greenhouse
m_{trans}	water net flow from transpiration of plants	X_i	indoor CO ₂ concentration level
m_{vent}	water net flow from ventilation	X_{inj}	control input of CO ₂ injection
P	atmospheric pressure	X_v	CO ₂ net flow from ventilation
p_{sat}	saturated atmospheric pressure	X_{pho}	net photosynthesis rate
q_{cover}	heat flux through the cover	x_k	system states
q_l	heat flux from supplemental lighting	ρ	air density
q_{pd}	heat flux from pad	ρ_w	water density
q_p	heat flux from the heating pipe	τ_l	heat conversion percentage
q_s	net solar radiation	τ_s	shading percentage

controlled environment agriculture (CEA). Greenhouses, a key CEA technology, require substantial energy for heating, particularly in colder climates, where heating costs are a major operational burden [6]. By integrating bitcoin mining waste heat into greenhouse climate control, this system can reduce reliance on fossil fuels, lower greenhouse gas emissions, and enhance the economic sustainability of food production. In doing so, it addresses two critical global challenges: reducing the environmental impact of energy-intensive industries and enhancing sustainable food production systems.

Greenhouses, particularly in colder climates, require considerable energy for heating, which contributes to the high operational costs and carbon footprint of agricultural production [7,8]. By harnessing the excess heat from bitcoin mining operations, greenhouses can potentially reduce their reliance on traditional energy sources, thereby lowering costs and minimizing environmental impact. A similar waste heat recovery system can also be observed in building integrated greenhouses [9,10]. This symbiotic relationship offers a promising approach to sustainable agriculture, where the convergence of digital and agricultural technologies could lead to new economic models and more efficient resource use. The exploration of such integrated systems is not only timely but necessary, given the increasing global emphasis on sustainability and energy efficiency.

Bitcoin mining is widely recognized for its intensive energy consumption. Global mining operations are estimated to consume over 170 TWh of electricity annually [11], an amount comparable to the total energy usage of some nations [12], with nearly all of this energy ultimately converted into waste heat. While efforts to reduce bitcoin mining's environmental footprint have primarily focused on transitioning to renewable energy sources, effectively utilizing the excess heat remains a largely unaddressed challenge. Recent research suggests that this heat could be repurposed, reducing waste and improving the overall efficiency of mining operations [5].

Greenhouse farming is a vital sector in agriculture, providing controlled environments for crop production, especially in regions with unfavorable climates [13]. The energy requirements for heating

greenhouses are substantial, and traditional energy sources like fossil fuels contribute to both high operational costs and greenhouse gas emissions. Studies have explored various alternative heating methods, such as geothermal energy [14], biomass, and solar power [15–17], to reduce the environmental footprint of greenhouses. However, these methods still involve significant upfront costs and logistical challenges.

The concept of utilizing waste heat from industrial processes is well-established, with various industries implementing this approach to improve energy efficiency. Case studies in Scandinavian countries have demonstrated the feasibility of repurposing waste heat from factories for district heating systems, providing both economic and environmental benefits to industries and local communities [18]. However, in the agricultural sector, the integration of digital technologies—particularly bitcoin mining—with greenhouse operations remains largely unexplored. Preliminary studies suggest that such integration would be economically profitable [5]. Beyond operational benefits, incentivizing the use of bitcoin mining waste heat for agricultural applications could drive policy changes aimed at reducing greenhouse gas emissions, promoting renewable energy adoption, and enhancing food security in energy-intensive agricultural systems.

Recent publications have reported various greenhouse climate control studies, such as model predictive control (MPC) [19], energy balance models [20], robust control [21], nonlinear control approaches [22], optimal control [23], and adaptive control [24]. MPC is a potent strategy to control the system by predicting future states from a model and disturbance predictions. MPC can predict future system states and make decisions that optimize energy use and greenhouse conditions over a given horizon [25]. Some applications of MPC in intelligent agriculture include irrigation [26] and greenhouse indoor climate [27]. MPC is also adopted in several studies regarding building climate control [28,29]. MPC is well-suited for greenhouse climate control, as the dynamics of greenhouse can be captured by first principle models, and the disturbance prediction can be easily gathered from weather forecasts [30]. Some research on greenhouse climate control has already demonstrated the advantages of MPC over traditional approaches, such

as proportional–integral–derivative (PID) control [25,31,32] and On-Off control [19,22,33–35]. Some prior research on greenhouse climate control employed a linear MPC approach [36,37]. In order to handle weather forecast errors and provide a robust controlled environment in a greenhouse, machine learning algorithms could be leveraged and incorporated in a data-driven robust MPC framework [6,13]. The nonlinear relationships between each system state in a greenhouse indicates that nonlinear MPC (NMPC) is an applicable method for addressing nonlinearity within greenhouse climate control [38]. Many NMPC-based studies solely consider temperature control in their control frameworks [33,35,39]. In [19], NMPC with genetic algorithms was adopted to regulate greenhouse temperature and humidity. However, CO₂ level control is not included. In [40,41], temperature, humidity, and CO₂ concentration were controlled, but a lighting system was not considered in the model. In [42], the four important system states were included in the model, but the lighting system was not adjusted according to the daily sunlight intensity and was scheduled in advance. Therefore, NMPC presents a viable control strategy for the energy management of the greenhouse climate but still requires the control framework capable of integrating the greenhouse and the data miners seamlessly.

Integrating bitcoin mining with greenhouse operations presents significant challenges, primarily in achieving precise control over the greenhouse climate to support optimal plant growth. The steady waste heat generated by mining hardware provides a reliable energy source, but effectively utilizing it requires advanced control strategies. Sophisticated systems must balance the greenhouse's thermal requirements with external weather conditions and internal crop needs, ensuring that heat distribution aligns with the dynamic demands of greenhouse operations.

This study develops and evaluates an NMPC framework for optimizing the integration of bitcoin mining waste heat into greenhouse operations to enhance energy efficiency, crop productivity, and economic viability. The nonlinear dynamic models that describe greenhouse climate variables—temperature, humidity, and CO₂ concentration—are initially developed. These models are then incorporated into the NMPC framework, which iteratively solves a nonlinear programming problem to determine optimal control inputs, such as fans, blinds, heating systems, and CO₂ injection. A simulated case study of a bitcoin mining integrated greenhouse is conducted to assess the NMPC framework's effectiveness under diverse environmental conditions. The case study considers three greenhouse sizes—individual, semi-commercial, and commercial—across eight cities representing various climate zones. This evaluation tests the NMPC framework's ability to optimize climate control while utilizing bitcoin mining waste heat, ensuring energy-efficient operation and stable growing conditions.

The remainder of this paper is structured as follows: Section 2 presents the dynamic models for greenhouse temperature, humidity, and CO₂ concentration, followed by the NMPC framework designed for optimizing greenhouse climate control with bitcoin mining waste heat. Section 3 examines the case study simulation outcomes to assess the effectiveness of the proposed control framework. Lastly, Section 4 presents a summary of the main findings and final conclusions.

2. Bitcoin mining integrated greenhouse dynamic climate models formulation

Fig. 1 shows control framework employed to maintain optimal climate conditions in a greenhouse integrated with bitcoin mining hardware. In this setup, key greenhouse climate states—such as temperature, humidity, CO₂ levels, and light intensity—are continuously monitored. These states serve as inputs to the NMPC framework, which dynamically processes this information to make real-time adjustments that ensure ideal conditions for plant growth. The MPC framework controls various actuators, including fans, blinds, CO₂ injection, and lighting, to modulate the greenhouse environment as needed.

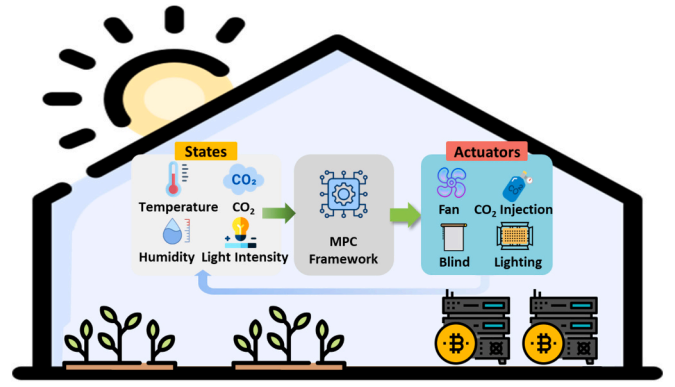


Fig. 1. Overview of the CO₂, moisture, and energy exchange dynamics between the building and an i-RTG. Actuator inputs are derived from the MPC controller, and key system variables like CO₂ levels, humidity, and temperature are under constant surveillance.

Additionally, the waste heat produced by bitcoin mining hardware contributes to heating the greenhouse, supporting energy efficiency and enhancing sustainability in this integration.

2.1. Bitcoin mining greenhouse climate dynamic model formulation

This study investigates the internal climate states of the bitcoin mining greenhouse, specifically temperature, humidity, and CO₂ levels, using a nonlinear dynamic model that captures their interactions with control inputs and external disturbances. This model is used to predict the greenhouse's future climate conditions, which is essential for minimizing control costs and preventing unfavorable environments for plant growth within the NMPC framework. Extreme temperatures can hinder plant development, while maintaining optimal CO₂ levels boosts photosynthesis and crop yields. However, excessive humidity can promote mold growth, adversely impacting crop quality and yield. Therefore, precise constraints are necessary to ensure plant health.

Key influencers of the bitcoin mining integrated greenhouse's temperature include heating pipe q_p , ventilation q_v , waste heat from supplemental lighting q_l , pad cooling q_{pd} , heat flux through the cover q_c , solar radiation q_s , and waste heat from bitcoin mining hardware q_b . The continuous-time dynamic model for temperature is expressed as [42]:

$$\rho V C_a \frac{dT_i}{dt} = q_s + q_p + q_l + q_b - q_c - q_v - q_{pd} \quad (1)$$

where ρ , V , and C_a denote air density, greenhouse volume, and the specific heat capacity of air. Eq. 1 describes a transient thermal energy balance for the greenhouse air, where the temperature is modeled as a dynamic state variable evolving over time due to net heat fluxes from heating systems, ventilation, lighting, solar radiation, pad cooling, and heat exchange through the greenhouse envelope. The model assumes a lumped thermal mass approach with spatially uniform air temperature, consistent with typical greenhouse climate control studies [43]. This formulation enables integration into the NMPC framework while maintaining computational efficiency.

$$q_s = A_s I_o K_a (1 - \tau_s u_{blind}) \quad (2)$$

$$q_p = \rho_w C_w K_h T_h u_{pipe, max} u_{pipe} \quad (3)$$

$$q_l = \tau_l u_{light, max} u_{light} \quad (4)$$

$$q_c = A_s K_c (T_i - T_o) \quad (5)$$

$$q_v = \rho C_a u_{fan, max} (T_i - T_o) u_{fan} \quad (6)$$

$$q_{pd} = A_p K_p u_{pad, max} (T_p - T_i) u_{pad} \quad (7)$$

where ρ , ρ_w , V , C_a , C_w , I_o , τ_s , and A_s represent density of air and water, greenhouse volume, specific heat capacity of air and water, outdoor global radiation, shading percentage, and greenhouse surface area, respectively. K_a , K_c , K_h , and K_p are coefficients of blind, cover, heating pipe, and pad cooling, respectively. The control inputs include pad shutter, shade curtain position, fan speed, and heating pipe water flow rate, represented by u_{pad} , u_{blind} , u_{fan} , and u_{pipe} . τ_l , A_p , T_h , T_o , and T_p represent the heat conversion percentage, pad area, temperatures of heating pipe water, ambient air, and air through the cooling pad, respectively.

The heat energy model for bitcoin mining machine is shown as follows [44]:

$$q_b = m_s C_a (T_{b,i} - T_{b,o}) \quad (8)$$

$$m_s = \rho V_b \quad (9)$$

where m_s is the air mass flow rate through the bitcoin mining machine, $T_{b,i}$, $T_{b,o}$ denotes the air temperature of inlet and outlet, and V_b represents the volume flow rate generated by the bitcoin mining machine fans.

For the dynamic humidity model of the bitcoin mining greenhouse, we identify the main contributors in the model. Relative humidity is influenced by both temperature and absolute humidity, with absolute humidity serving as the foundation of the mass balance equation. The mass balance for absolute humidity considers the net water flows from ventilation m_{vent} , plant evapotranspiration m_{trans} , and the fogging system m_{fog} , expressed as follows:

$$\rho V \frac{dh_i}{dt} = m_{vent} + m_{trans} + m_{fog} \quad (10)$$

The CO₂ level is a key factor affecting photosynthesis rates in plants, enhancing yield and quality when increased to a specific threshold. The CO₂ mass balance takes into account the internal CO₂ concentration X_i , net CO₂ from ventilation X_{vent} , consumption from photosynthesis X_{pho} , and CO₂ injection X_{inj} , as shown below:

$$\rho V \frac{dX_i}{dt} = X_{vent} - X_{pho} + X_{inj} \quad (11)$$

The dynamic models of temperature, humidity, and CO₂ concentration described above are embedded in the NMPC framework to predict future climate states and determine optimal control inputs over a moving prediction horizon.

2.2. Nonlinear model predictive control framework for bitcoin mining integrated greenhouse climate

Waste heat from bitcoin mining hardware is incorporated into the greenhouse model. The NMPC framework manages the climate within the bitcoin mining greenhouse, aiming to minimize energy and CO₂ costs for the greenhouse operation.

NMPC is shown to be effective for climate control within the integrated greenhouse, utilizing the nonlinear dynamic models in a compact format where H represents the forecast horizon. This compacted model consists of the nonlinear vector function $\phi(\bullet)$, control inputs $\mathbf{u}$, sequences of disturbances $\hat{\mathbf{v}}_k$, initial states of the system $\hat{\mathbf{q}}_k$, and the current system state $\mathbf{x}$. The model is expressed as follows:

$$\mathbf{x} = \phi(\hat{\mathbf{q}}_k, \mathbf{u}, \hat{\mathbf{v}}_k) \quad (12)$$

The NMPC framework integrates the nonlinear dynamic model along with necessary constraints. Constraints on the control inputs and system states are defined by $\mathbf{F}_u \mathbf{u} \leq \mathbf{f}_u$ and $\mathbf{F}_x \mathbf{x} \leq \mathbf{f}_x$, where $\mathbf{F}_x$, $\mathbf{f}_x$, $\mathbf{F}_u$, and $\mathbf{f}_u$ determine the limits for both control inputs and system states. The nonlinear optimization problem within this context is presented as follows:

$$\begin{aligned} \min_{\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\varepsilon}} J_k &= \mathbf{c}^T \mathbf{u}_k + \boldsymbol{\varepsilon}^T \mathbf{S} \boldsymbol{\varepsilon} \\ \text{s.t. } \mathbf{x}_k &= \phi(\hat{\mathbf{q}}_k, \mathbf{u}_k, \hat{\mathbf{v}}_k) \\ \mathbf{F}_x \mathbf{x}_k &\leq \mathbf{f}_{x,k} + \boldsymbol{\varepsilon} \\ \mathbf{F}_u \mathbf{u}_k &\leq \mathbf{f}_{u,k} \end{aligned} \quad (13)$$

where $\mathbf{c}$, $\mathbf{S}$, and $\boldsymbol{\varepsilon}$ denote cost coefficients, constraint violation penalty weight matrix, and slack variables.

Fig. 2 shows the control framework of the proposed NMPC framework for bitcoin mining integrated greenhouse climate. The framework operates using a receding horizon strategy. At each time step, weather forecasts and system state data are gathered. The optimal control actions are then determined by solving a nonlinear optimization problem. From

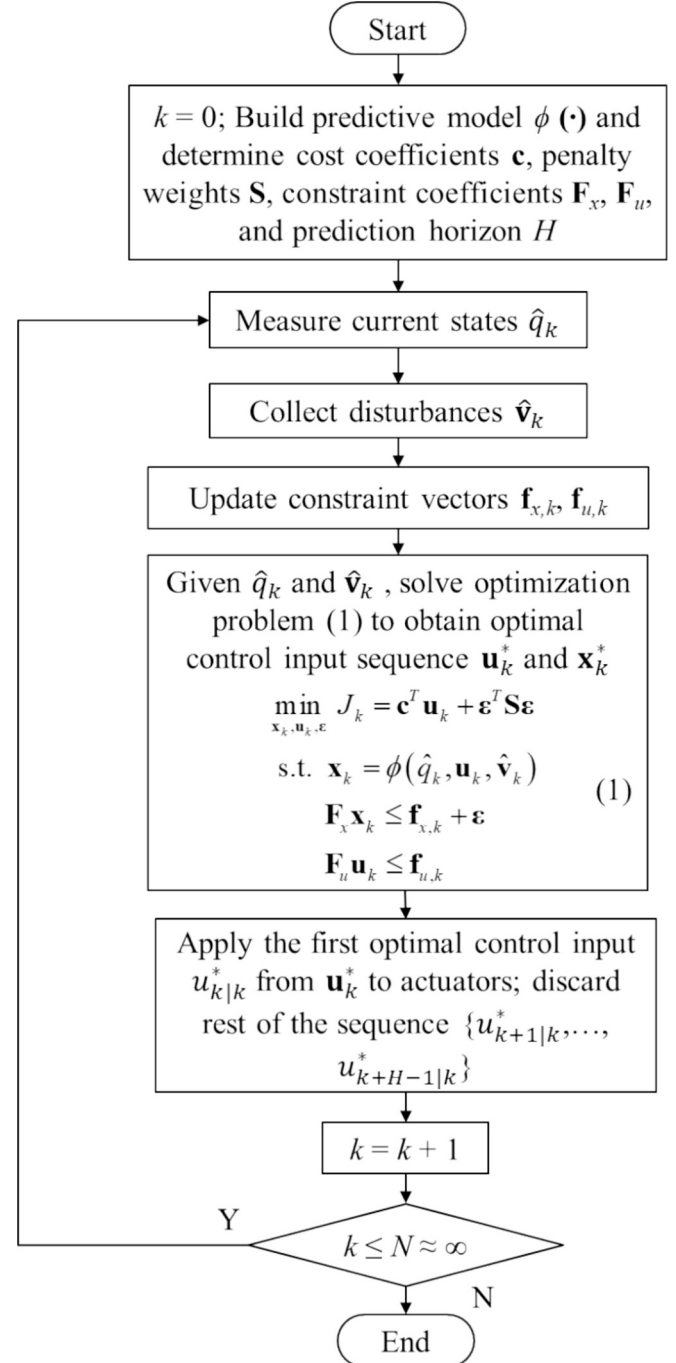


Fig. 2. The NMPC framework for bitcoin mining integrated greenhouse climate.

the computed sequence of control inputs, only those corresponding to the current time step are implemented, while the process repeats for subsequent steps. The NMPC framework regulates the climate of the bitcoin mining-integrated greenhouse by transmitting the control inputs to actuators such as fans, blinds, lighting, and CO₂ injection systems.

3. Case study of bitcoin mining integrated greenhouses

3.1. Bitcoin mining integrated greenhouse descriptions

In this study, climate control simulations are conducted for bitcoin mining greenhouses located in eight cities (See Fig. 3). This study is simulation-based and does not involve a physical experimental setup. The greenhouse climate is modeled using nonlinear dynamic equations, and NMPC is used to optimize the operation of climate control actuators. Standard actuators—including fans, pad cooling, CO₂ injection, and supplemental lighting—are incorporated into the model. The simulations cover a 60-day period for each season, starting on January 1st, April 1st, July 1st, and October 1st of 2021. Weather forecasts are used to address the optimal control problem, with the NMPC strategy updating every 15 min and employing a control horizon of 6 h. At each time step, the system state is updated based on weather data and predictive models. On-Off control is also implemented for comparison with the NMPC strategy. While no experimental system is implemented, the modeling assumptions and parameter values are based on literature and typical operational conditions. Different greenhouse sizes are tested, with the original size classified as a semi-commercial greenhouse and the commercial size having a width four times larger than the semi-commercial size.

Antminer S19 Pro bitcoin miners, each with a power consumption of 3.25 kW, are used in the simulation [45]. This study assumes that the bitcoin mining hardware is located within the greenhouse structure, and the waste heat generated is directly transferred into the greenhouse air as a continuous internal heat source. This setup eliminates the need for external heat capture or ducted distribution, allowing for immediate utilization of the thermal energy produced by mining equipment. This co-location strategy is consistent with prior research [5], which demonstrated the effectiveness of bitcoin miner-heated greenhouses. Although thermal storage is not included in this model, the system assumes stable mining operation as the primary heat source. Future investigations may explore the integration of heat buffering systems to accommodate operational fluctuations and further enhance temperature regulation.

Table 1 presents the optimal climate conditions for a bitcoin mining integrated greenhouse, listing target ranges for temperature, relative humidity, CO₂ concentration, and light intensity to support optimal plant growth. Maintaining these conditions is crucial for enhancing crop yield and quality within the greenhouse environment. The temperature range is set to balance heating from bitcoin mining waste heat with the cooling needs to avoid plant stress, while relative humidity is optimized

Table 1

Optimal climate conditions for the bitcoin mining integrated greenhouse.

	Optimal Climate Conditions
Temperature [°C]	Light period: 21.5–24 Dark period: 17.5–20
Relative humidity [%]	50–70
CO ₂ concentration [ppm]	Light period: > 1000

to prevent excessive moisture that could lead to mold or fungal issues. The CO₂ concentration level is increased within a controlled range to boost photosynthesis and maximize productivity. Light intensity is also carefully regulated, ensuring that plants receive adequate illumination without excessive energy use. These optimal climate parameters serve as reference points for the NMPC framework, which dynamically adjusts the greenhouse environment by utilizing bitcoin mining waste heat alongside traditional climate control elements. This table highlights the ideal setpoints the NMPC strives to maintain, aiming to create a stable, resource-efficient, and productive greenhouse climate.

Temperature domains categorized in [46] are used in this study. The six temperature zones are tropical, subtropical, warm temperate, cool temperate, boreal, and polar. Additionally, each location falls into one of three moisture categories: desert, dry, and moist. Cities in countries where bitcoin mining is restricted, such as China, are excluded from this study. The climate zones and moisture categories for the eight selected cities are presented in Table 2.

3.2. Results and discussion

All climate profiles, violation metrics, and operating cost analyses presented in the Results section are obtained by simulating the NMPC framework with the nonlinear dynamic models described in Section 2. The simulations are performed over a 60-day period for each season and location, with the models solved iteratively at 15-min intervals. All computations are executed using MATLAB on a machine equipped with an Intel Core i7–6700 CPU (3.40 GHz) and 32 GB of RAM.

Table 2

Climate zones and moisture domains of eight different cities. (Zone 1: Tropical, Zone 2: Subtropical, Zone 3: Warm temperate, Zone 4: Cool temperate, Zone 5: Boreal, Zone 6: Polar).

Cities	Climate Zones	Moisture Domains
Kuala Lumpur, Malaysia	1: Tropical	Moist
Phoenix, Arizona	2: Subtropical	Dry
Tokyo, Japan	3: Warm temperate	Moist
Columbus, Ohio	3: Warm temperate	Moist
Berlin, Germany	3: Cool temperate	Moist
Ithaca, New York	4: Cool temperate	Moist
Stockholm, Sweden	5: Boreal	Moist
Reykjavik, Iceland	6: Polar	Moist



Fig. 3. Eight cities of the case study with the climate zone information.

Fig. 4 presents a comparison of temperature control profiles between NMPC and On-Off control over a 14-day period in winter, spring, summer, and fall, respectively. The simulations are part of a 60-day study for each season, beginning on January 1st, April 1st, July 1st, and October 1st of 2021. In the figure, NMPC is represented by solid lines, while On-Off control is depicted by dotted lines. The dashed lines indicate the upper and lower temperature bounds.

The results show that NMPC consistently maintains the temperature within the specified bounds across all seasons, achieving smoother and more stable control compared to On-Off control. On-Off control, by contrast, produces significant oscillations in the temperature profile, frequently breaching the upper and lower bounds. This is particularly evident during periods of rapid environmental changes, where the reactive nature of On-Off control struggles to adapt. The temperature stability achieved by NMPC had a standard deviation of ± 1.3 °C compared to ± 3.9 °C for On-Off control, indicating significantly better performance. The predictive and adaptive capabilities of NMPC allow it to anticipate changes and optimize control actions, resulting in better overall climate stability for the bitcoin mining integrated greenhouse. This highlights the effectiveness of NMPC in managing dynamic and

seasonal variations in temperature.

Table 3 compares the control performance of On-Off control and NMPC across various metrics: control cost, cost savings, violation percentage, and violation amount. Control costs include electricity expenses for fans, pad cooling, shade curtains, lighting, and CO₂ enrichment. The data shows that NMPC consistently achieves lower control costs than On-Off control, resulting in significant cost savings

Table 3

Performance comparison of On-Off control and NMPC in bitcoin mining greenhouse climate control. Control costs include electricity for fans, pad cooling system, shade curtains, lighting system, and CO₂ cost.

	On-Off	NMPC
Control cost (USD)	6438	5586
Cost savings	–	13.2 %
Temperature violation percentage	37.2 %	19.7 %
Violation amount (°C•h)	659	103
Humidity violation percentage	57.3 %	35.2 %
CO ₂ violation percentage	28.7 %	14.9 %

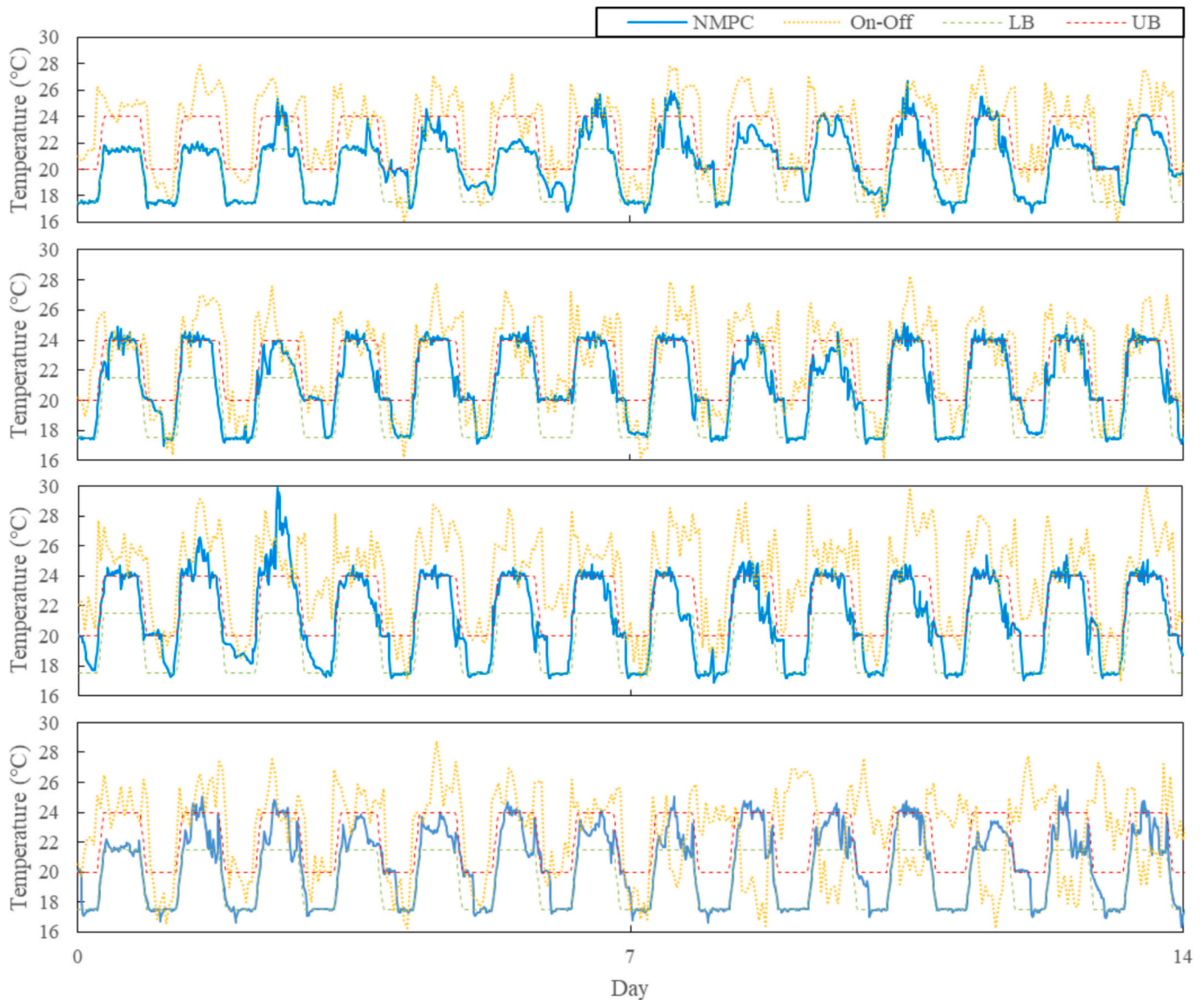


Fig. 4. Temperature profiles comparison between NMPC and On-Off control for 14 days in winter, spring, summer, and fall, respectively. The NMPC and On-Off control profiles are represented by the solid and dotted lines. The upper and lower bounds are shown in dashed lines. This figure shows deterministic simulation results based on fixed weather and system parameters; therefore, no error bars are included.

across different seasonal conditions. The cost savings highlight NMPC's efficiency in optimizing energy and resource use, particularly through its ability to predict and adapt to varying climate conditions within the greenhouse. Additionally, NMPC exhibits a much lower violation percentage and violation amount compared to On-Off control, meaning it better maintains the climate within desired temperature and CO₂ levels.

This reduction in violations aligns with the temperature profiles previously shown, where NMPC maintained a stable temperature within the upper and lower bounds, while On-Off control experienced frequent boundary violations. These results demonstrate that NMPC provides a more cost-effective and reliable climate control solution for bitcoin mining greenhouses, minimizing both operational costs and deviations from optimal growing conditions.

The annual operating costs of climate control activities—lighting, cooling, heating, and CO₂ enrichment—per square meter for bitcoin mining greenhouses at eight locations: Kuala Lumpur, Phoenix, Tokyo, Columbus, Berlin, Ithaca, Stockholm, and Reykjavik are illustrated in Fig. 5. The cost components are broken down to reveal the relative contributions of each activity. The results highlight the variation in operating costs across different climatic regions. Locations with colder climates, such as Reykjavik, Ithaca, and Stockholm, exhibit higher heating costs due to the greater demand for maintaining optimal greenhouse temperatures. Conversely, in warmer regions like Phoenix and Kuala Lumpur, cooling costs are significantly higher, reflecting the effort to mitigate excessive heat. Lighting costs are relatively stable across most locations, while CO₂ enrichment costs contribute a smaller but consistent portion of the total operating expenses. The NMPC framework plays a crucial role in optimizing resource allocation and reducing operational costs by dynamically adjusting climate control strategies based on real-time environmental conditions and energy demands. For example, in colder regions such as Stockholm and Reykjavik, NMPC reduced heating energy consumption by approximately 15 % compared to traditional methods, demonstrating its efficiency in optimizing heat utilization and minimizing energy waste. By leveraging predictive control, NMPC balances energy inputs across heating, cooling, and ventilation, ensuring stable indoor conditions while reducing overall operating expenses. This breakdown underscores the influence of local climate conditions on energy usage and operational expenses, providing valuable insights for optimizing resource allocation and improving the economic feasibility of bitcoin mining greenhouses in diverse environments.

Fig. 6 compares the energy consumption per square meter of individual, semi-commercial, and commercial-sized bitcoin mining greenhouses across eight locations. The energy consumption reflects the combined demands for heating, cooling, lighting, and other climate control activities necessary to maintain optimal growing conditions.

The results indicate clear variations based on location and greenhouse size. Colder climates, such as Reykjavik, Stockholm, and Ithaca, demonstrate higher energy demands due to the significant heating requirements for maintaining optimal greenhouse conditions. In warmer

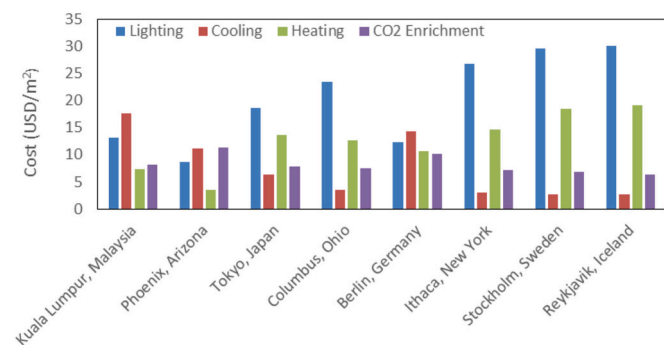


Fig. 5. Breakdown of annual operating costs for climate control activities in bitcoin mining greenhouses across eight locations.

regions like Kuala Lumpur and Phoenix, energy consumption is lower but remains substantial, likely driven by cooling needs. Notably, semi-commercial greenhouses exhibit slightly higher energy consumption per square meter compared to their commercial counterparts in most locations, which could be attributed to differences in scale efficiency and operational optimization. This analysis underscores the importance of tailoring energy strategies to both local climate conditions and greenhouse size. The results emphasize the need for efficient energy management systems, particularly in bitcoin mining integrated greenhouses, to reduce operational costs and promote sustainable practices.

The monthly natural gas expenses required for heating individual, semi-commercial, and commercial greenhouses across eight locations is shown in Fig. 7. The result highlights the significant variation in heating costs throughout the year, with expenses peaking during winter months (January, February, and December) due to increased heating demands in colder climates. Conversely, heating expenses are minimal or negligible during the summer months (June, July, and August), reflecting reduced heating requirements. As expected, larger greenhouse sizes result in higher heating expenses, with commercial greenhouses exhibiting the greatest costs, followed by semi-commercial and individual greenhouses. For instance, in January and December, the expenses for commercial greenhouses far exceed those of smaller systems, emphasizing the scaling effects of heating demands. The average monthly expenses also demonstrate the proportional relationship between greenhouse size and energy consumption. These results underscore the critical importance of integrating alternative heating sources, such as waste heat from bitcoin mining, to offset natural gas expenses and improve the sustainability and economic feasibility of greenhouse operations. Utilizing bitcoin mining waste heat could reduce dependence on conventional heating methods, particularly in colder months, thereby decreasing overall operational costs and carbon emissions.

Fig. 8 compares the annual heating expenses (blue bars) and potential profits (orange, gray, yellow, and light blue bars) for individual, semi-commercial, and commercial greenhouses across eight locations under four different bitcoin price scenarios: high, medium, low, and dynamic. The natural gas expenses represent the cost of conventional heating without bitcoin mining waste heat integration, while profits reflect the financial gains from utilizing bitcoin mining waste heat, offset by the prevailing bitcoin price. To illustrate the effectiveness of bitcoin mining waste heat integration, the heating expenses for conventional greenhouses operating without waste heat recovery is represented by the natural gas bars. These serve as a baseline to compare against the profit values under different bitcoin price scenarios. The comparison clearly shows that in cold-climate cities such as Reykjavik and Stockholm, natural gas expenses can exceed \$500,000 annually for a commercial greenhouse. In contrast, when mining waste heat is utilized, heating expenses are largely offset, resulting in net positive profits even under fluctuating bitcoin market conditions. This side-by-side comparison emphasizes the dual benefits of energy savings and increased profitability enabled by the integration of waste heat, particularly in high-demand regions. The results show that natural gas expenses remain consistent across locations and greenhouse sizes, with colder climates such as Reykjavik, Stockholm, and Ithaca incurring the highest costs due to greater heating demands. However, bitcoin mining waste heat significantly impacts profitability.

Under high bitcoin prices (orange bars), all greenhouse sizes across most locations yield substantial profits, particularly in colder regions where heating demands are greater. Conversely, with medium (gray bars) and low bitcoin prices (yellow bars), profitability varies widely, with colder locations still benefiting more due to higher heating cost offsets. A single Antminer S19 Pro bitcoin miner produces approximately 3.25 kW of thermal energy [44]. This integration significantly reduces the reliance on natural gas, particularly in colder regions. The offset in heating demand leads to substantial cost savings—up to 60 % reduction in annual heating expenses—contributing to net profits of up to \$1.5 million for commercial greenhouses under high bitcoin price

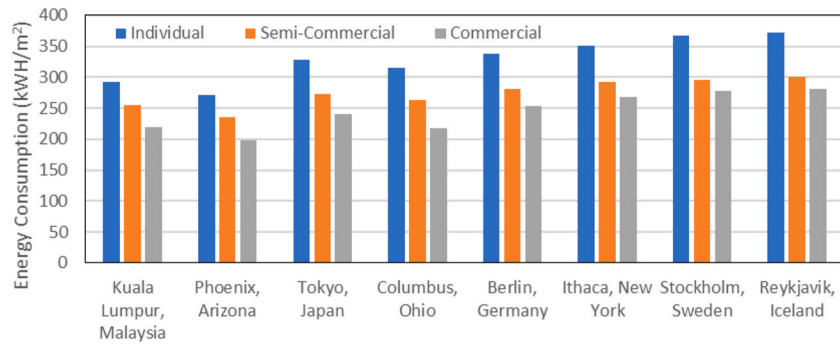


Fig. 6. Breakdown of annual energy consumption for climate control activities in bitcoin mining greenhouses across eight locations.

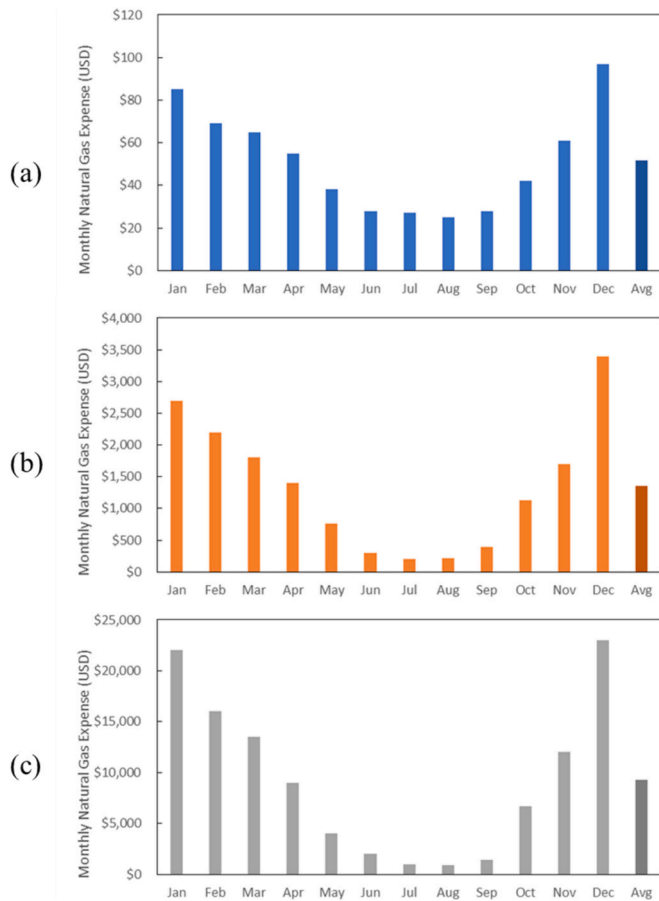


Fig. 7. Monthly natural gas expense to heat (a) individual, (b) semi-commercial, and (c) commercial greenhouses besides bitcoin mining in Ithaca, New York.

scenarios.

To enhance realism, Fig. 8 also includes an additional case using dynamic bitcoin prices and dynamic network difficulty (light blue bars), which reflect market fluctuations over the course of a year. The results show that even under these variable conditions, profitability remains positive across most locations, especially for commercial greenhouses in colder climates. However, the margins are generally lower compared to the high-price scenario, highlighting the financial uncertainty introduced by market volatility. This reinforces the need for adaptive strategies and flexible integration frameworks that can account for temporal changes in mining operations.

Commercial greenhouses exhibit the largest potential profits, reflecting their scale and greater heating requirements, while individual

greenhouses show the smallest margins. This highlights the scalability of integrating bitcoin mining with greenhouse operations. These findings underscore the economic viability of using bitcoin mining waste heat to offset heating expenses, particularly in colder regions and under favorable bitcoin market conditions. However, the dependency on bitcoin price introduces variability in profitability, emphasizing the need for adaptive strategies to optimize integration in diverse economic and climatic scenarios.

One limitation of this study is that it does not account for variability in bitcoin mining heat output, which could impact greenhouse climate stability. Fluctuations in mining activity—driven by market volatility, energy costs, or regulatory constraints—may result in inconsistent waste heat availability. This unpredictability could lead to temperature deviations, requiring additional energy inputs to maintain optimal growing conditions. Future work should incorporate models that account for these fluctuations to better assess their impact on climate control and operational costs. Additionally, integrating adaptive control strategies, such as NMPC, with supplementary energy sources like renewable energy or backup heating systems could help mitigate risks associated with heat variability, enhancing the resilience of bitcoin mining-integrated greenhouses.

This study adopts a lumped-parameter modeling approach to describe the thermal and climate dynamics of the greenhouse. While this approach abstracts away spatial variability such as nonuniform temperature distribution or airflow dynamics, it enables computationally tractable real-time optimization with MPC [43,47]. Incorporating spatially resolved models—such as CFD-based or zonal models—may offer greater physical accuracy but is computationally prohibitive for MPC applications. Future research could investigate reduced-order models that strike a balance between spatial fidelity and control feasibility.

4. Conclusions

This study developed and evaluated a NMPC framework for integrating bitcoin mining waste heat with greenhouse climate control. The proposed approach leverages real-time optimization to regulate temperature, humidity, and CO₂ concentration while minimizing operational costs. A simulated case study across eight global locations, considering individual, semi-commercial, and commercial greenhouse sizes, demonstrated that NMPC significantly enhances energy efficiency and economic viability. Results indicate that NMPC reduces energy consumption by up to 15 % and lowers operational costs by 20–30 % compared to conventional On-Off control. Furthermore, economic analysis highlights that commercial-scale greenhouses in colder regions can achieve net annual profits of up to \$1.5 million under favorable bitcoin market conditions, reinforcing the financial feasibility of repurposing mining waste heat for controlled environment agriculture.

Despite these promising findings, the study is subject to several assumptions and limitations. The analysis assumes consistent bitcoin

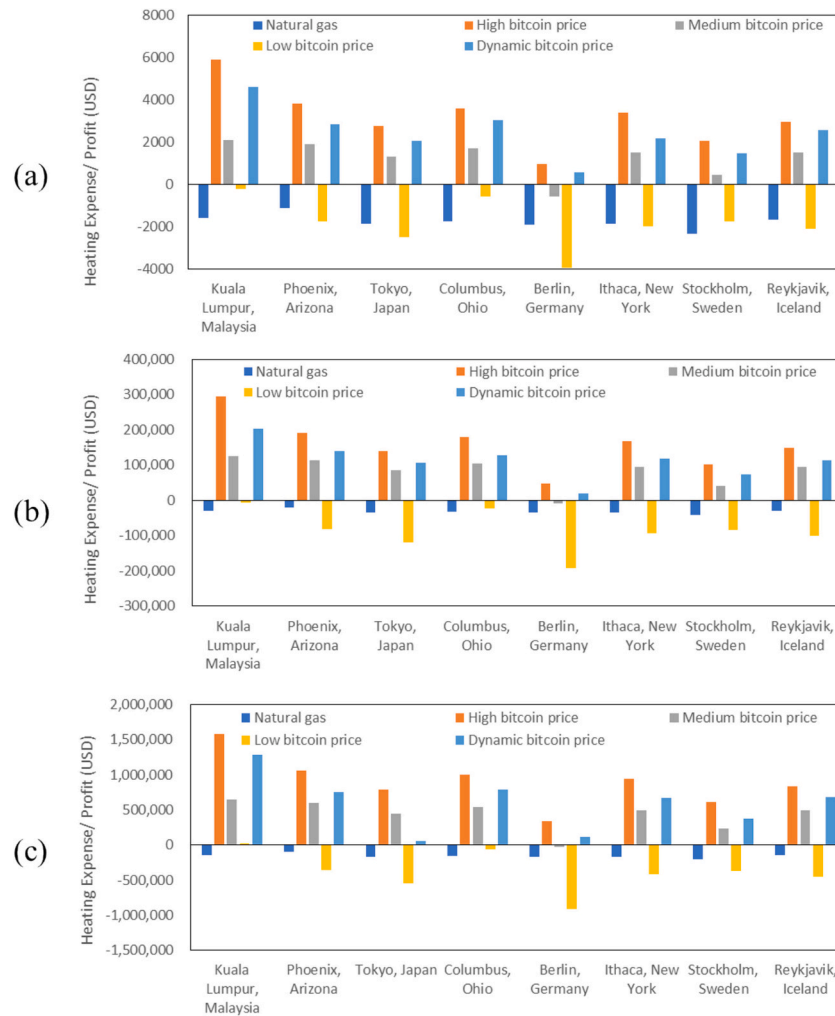


Fig. 8. Annual heating expense and profit for (a) individual, (b) semi-commercial, and (c) commercial greenhouses across eight locations.

mining operations and does not account for significant fluctuations in bitcoin prices or mining activity. In reality, mining operations may vary due to market volatility, energy costs, and regulatory constraints, leading to inconsistent waste heat availability. Future work should incorporate stochastic modeling to account for these uncertainties and assess their impact on greenhouse climate stability and profitability. Additionally, while NMPC provides real-time optimization, the framework does not currently integrate backup energy sources to compensate for heat variability. A hybrid approach incorporating renewable energy or auxiliary heating systems could improve greenhouse resilience in fluctuating conditions.

Moreover, the study considers idealized system dynamics and does not include the physical constraints of heat distribution within the greenhouse. Heat transfer efficiency may vary due to factors such as air circulation, equipment placement, and greenhouse design, which could influence actual performance. Future experimental validation in a physical greenhouse prototype will be crucial to refine the dynamic models and confirm the practical effectiveness of the proposed NMPC strategy.

CRedit authorship contribution statement

Wei-Han Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Fengqi You:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Methodology,

Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Fengqi You reports financial support was provided by National Institute of Food and Agriculture. F.Y. is an Associate Editor of Applied Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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